

1

a

$$\hat{M}_{MLE} = x_n$$

b

Given from the announcement:

$$E(\hat{M}_{MLE}) = \frac{n}{n+1}(M+1)$$

This is a biased estimator, as $\hat{M}_{ME} \leq M$ and will underestimate M .

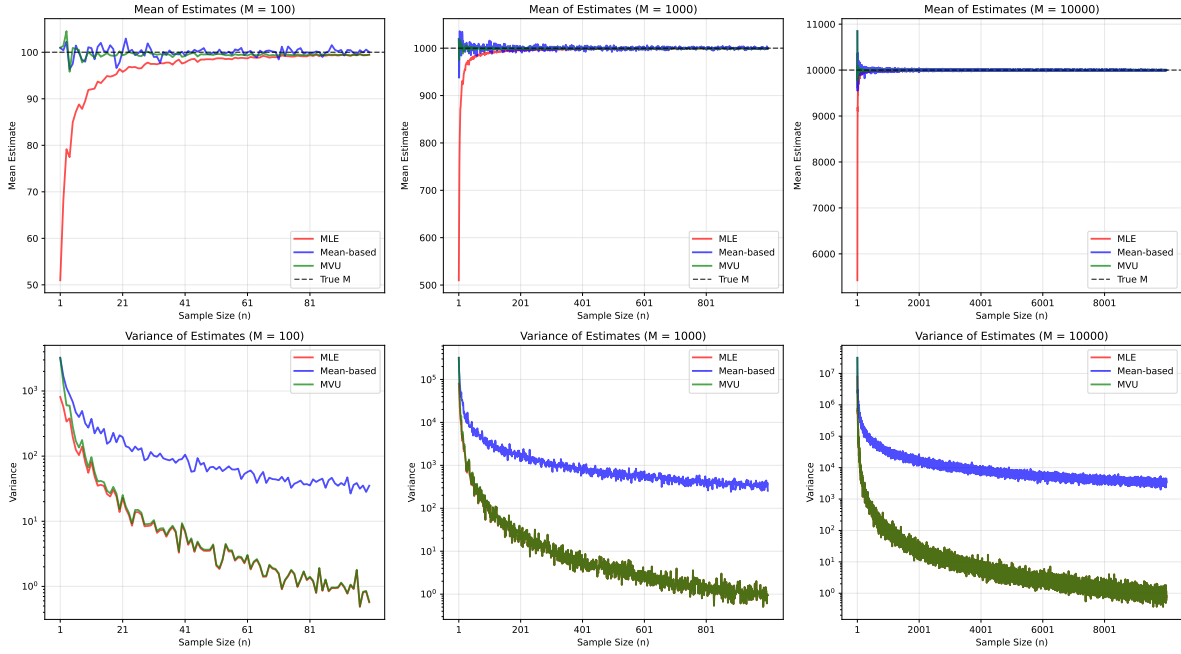
$$E(\hat{M}_{MEAN}) = M$$

This is an unbiased estimator as the expected value of M_{MEAN} is equal to the value it is trying to estimate (M)

$$E(\hat{M}_{MVU}) = M$$

This is an unbiased estimator as the expected value of M_{MVU} is equal to the value it is trying to estimate (M)

C



According to the Mean of estimates graph it is clear to see that the $\hat{M}_{MEAN}$ and $\hat{M}_{MVU}$ estimates are unbiased estimators of M as they their expected value is M . On the other hand, $\hat{M}_{MLE}$ is evidently an unbiased estimator as it falls significantly below M for smaller values of n . For sufficiently large n , the three estimators all approach M which is good.

Their variances on the other hand tell a different story. The $\hat{M}_{MEAN}$ estimator has an extremely volatile variance many orders of magnitude above both $\hat{M}_{MLE}$ and $\hat{M}_{MVU}$. This demonstrates that the $\hat{M}_{MVU}$ estimator should be the best estimator for M as the variance is tied for smallest and is an unbiased estimator of M .